

8.5 Quiz: Rational Functions and Transformations (no calculator)

1. [Maximum mark: 8]

Consider the function $f(x) = \frac{2}{x-3} + 1$.

- (a) Write down the equations of the vertical and horizontal asymptotes of the graph of f . [2]
- (b) Describe fully the sequence of transformations that maps the graph of $y = \frac{1}{x}$ onto the graph of f . [3]
- (c) Find the y -intercept of the graph of f . [1]
- (d) Find the x -intercept of the graph of f . [2]

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2. [Maximum mark: 8]

The function g is defined by $g(x) = \frac{2x+1}{x-1}$, $x \in \mathbb{R}$, $x \neq 1$.

(a) Write down the equations of the vertical and horizontal asymptotes of the graph of g . [2]

(b) Find the coordinates of the x -intercept and of the y -intercept of the graph of g . [2]

(c) Find $g^{-1}(x)$. [3]

(d) State the domain of g^{-1} . [1]

8.5 Quiz: Rational Functions and Transformations (no calculator)

1. [Maximum mark: 8]

Consider the function $f(x) = \frac{3}{x+2} - 1$.

(a) Write down the equations of the vertical and horizontal asymptotes of the graph of f . [2]

(b) Describe fully the sequence of transformations that maps the graph of $y = \frac{1}{x}$ onto the graph of f . [3]

(c) Find the y -intercept of the graph of f . [1]

(d) Find the x -intercept of the graph of f . [2]

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2. [Maximum mark: 8]

The function g is defined by $g(x) = \frac{3x-2}{x+1}$, $x \in \mathbb{R}$, $x \neq -1$.

(a) Write down the equations of the vertical and horizontal asymptotes of the graph of g . [2]

(b) Find the coordinates of the x -intercept and of the y -intercept of the graph of g . [2]

(c) Find $g^{-1}(x)$. [3]

(d) State the domain of g^{-1} . [1]

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8.5 Quiz: Rational Functions and Transformations (no calculator)

1. [Maximum mark: 8]

Consider the function $f(x) = \frac{4}{x-1} + 2$.

- (a) Write down the equations of the vertical and horizontal asymptotes of the graph of f . [2]
- (b) Describe fully the sequence of transformations that maps the graph of $y = \frac{1}{x}$ onto the graph of f . [3]
- (c) Find the y -intercept of the graph of f . [1]
- (d) Find the x -intercept of the graph of f . [2]

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2. [Maximum mark: 8]

The function g is defined by $g(x) = \frac{4x+3}{x-2}$, $x \in \mathbb{R}$, $x \neq 2$.

(a) Write down the equations of the vertical and horizontal asymptotes of the graph of g . [2]

(b) Find the coordinates of the x -intercept and of the y -intercept of the graph of g . [2]

(c) Find $g^{-1}(x)$. [3]

(d) State the domain of g^{-1} . [1]